

# ML Challenge 2026: Business Entity Resolution Solution Template

**Team Name:** Grenuke  
**Team Members:** Ameya Borkar, Aarush Bakshi, Sachi Dhoka  
**Submission Date:** 29 September 2026

**At a glance. Public leaderboard: 0.990879** macro F0.5, the official score of this package's submission ("Composite B"). **Local validation: 0.9913**, on our labelled US/India holdout (France has no labels). Blocking keeps **99.1%** of true pairs at **3.70 candidates per S1**. Only the provided data; every model is **MIT or Apache-2.0** and  $\leq 8B$  parameters.

## Contents

1. Executive Summary

2. Methodology

3. Candidate Generation (Blocking)

4. Matching Model

5. Results & Error Analysis

6. Conclusion

Appendix A. Code Artefacts

Appendix B. Additional Results

## 1. Executive Summary

We **retrieve** candidates, **score** every pair, and **decide** for each Source-1 (S1) business which set of Source-2/3 copies maximises the expected F0.5.

**Blocking** searches each country separately, and a first classifier trims its output to 3.70 candidates per S1. **Gradient-boosted classifiers** (XGBoost) then score each pair on name, address and legal-form evidence. For the pairs those classifiers are unsure about, fine-tuned multilingual **cross-encoders** read the text itself: e5, bge, and LoRA-tuned **Qwen2.5-1.5B and 7B**. A **set-level decision** finally chooses, for every S1, the copies that maximise expected F0.5 (Figure 1).

The hardest part was **France**, which appears only in the test set. We taught the models French through **guarded self-training** on our own decisions, and an **independent Qwen2.5-7B re-check** of the predictions that looked certain removed French look-alike decoys that string features accept.

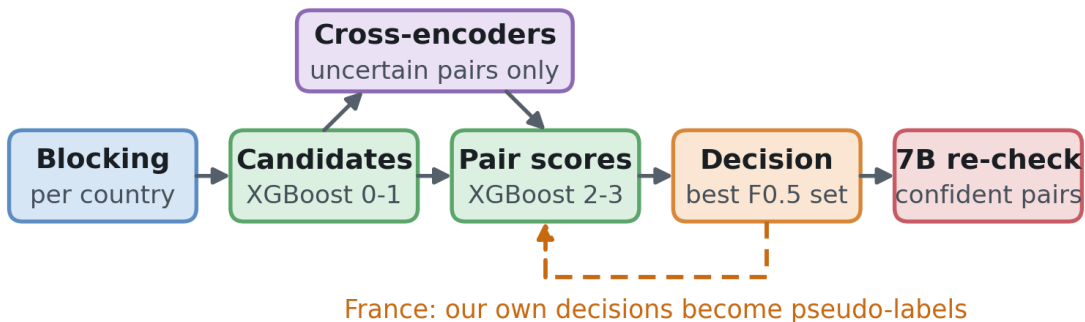


Figure 1. The pipeline, from records (left) to matches (right). The dashed loop is the French self-training (§2.2).

## 2. Methodology

---

*Three properties of the data shaped the design: a precision-heavy metric scored per S1, look-alike decoys, and a country without labels.*

### 2.1 Problem Analysis

**A precision-heavy, per-S1 metric.** Each S1 business has 0–11 noisy vendor copies, 3.46 on average. About 5.6% have none, and for those any prediction scores 0. F0.5 weighs precision twice as much as recall, so **a wrong merge costs more than a missed copy**. We therefore designed for precision first, and we decide on each S1's **whole set** rather than pair by pair.

**How copies differ.** On labelled pairs, true copies:

- change case and accents;
- drop or reformat **legal forms** (LLC, Pvt Ltd; SARL/SAS/EURL in about 23% of French copies);
- reorder or append words;
- contain **acronyms**, typos and garbles;
- abbreviate **street types** (Rue → R.) or reorder the address;
- change the **house number**, or leave the address **empty**.

Our blocking keys (§3) and features (§4) are built around exactly these edits.

**Look-alike decoys.** The test set has **about 23% more records per S1** than train, while the number of true matches per S1 stays flat. So the extra records are **decoys**: a business word swapped at the same address, a nudged house number, or the same name somewhere else. Because **46–54% of S1 share their name** with another S1, the address usually decides. So every record is owned by at most one S1, and candidates compete within each S1.

**A country without labels.** France appears only in the test set.

- **Generic names:** city plus category, e.g. "lille club sarl". Each French decoy we removed shares its name with a median of 43 other S1.
- **Shared addresses:** 11% of French S1 share their exact address with another S1.
- **Administrative names:** 32% of French addresses carry department or region names.

Our first uploads scored 0.976–0.980 on the public leaderboard, against 0.984–0.989 on our local US/India holdout. That implied France near 0.93, **so most of our later work targeted France**.

### 2.2 Solution Strategy

**Approach Type:** Hybrid. Multi-view blocking → gradient-boosted pair classifiers → cross-encoders on the uncertain pairs → a set-level decision → rules → an LLM re-check (Figure 1).

**Core Innovation:**

1. **Guarded, cross-fitted self-training for a country without labels.** We pseudo-label French pairs from our best model's final decisions: *positive* if kept with calibrated probability  $\geq 0.9$ , *negative* if rejected with  $\leq 0.05$ , and unlabelled otherwise. **Guards** protect against known failure modes: rule-derived populations override the labels, and empty-

address records keep the first round's labels. The labels are **cross-fitted by S1 group**, so no French pair is ever scored by a model that saw its own label. Round 1 added **+0.00046** on the leaderboard; round 2, together with the French decision layers, added **+0.00015**.

2. **A decision that optimises the metric itself.** Instead of one global threshold, each S1 gets the prediction set with the highest **expected F0.5** under calibrated probabilities, found by a small dynamic programme (§4). It beat a tuned threshold on our local holdout by **+0.000048** (95% CI +0.000007 to +0.000091).
3. **An independent second opinion on "certain" predictions.** 94.5% of final predictions have a stage-1 probability above 0.99, so no cross-encoder ever read them. A LoRA-tuned **Qwen2.5-7B** re-reads them and drops those it rejects strongly (logit < -6), a cut-off we fixed on labelled data beforehand. It removed **840 French predictions**; on labelled data, only 8% of such rejects are true matches.

**How we decided what to keep.** For US/India, our **local validation** is a **fixed holdout of 25% of train S1** (549,699 S1). It was never used for fitting, and we scored it with the official macro F0.5. A component was kept only if a **paired bootstrap** showed a gain, with ties going to the simpler option. Post-hoc rules also had to gain on **both halves** of the holdout. France has no labels, so we judged it with label-free checks, estimates corrected with the 7B's judgement, and leaderboard probes that changed only France.

### 3. Candidate Generation (Blocking)

*Blocking keeps 99.1% of true pairs while cutting the search to 3.70 candidates per S1.*

**Blocking keys used.** Every search runs **per country**, so IDF weights and neighbours are never borrowed across countries. Records with an empty country join every partition. Three complementary views each catch a different kind of noise (Table 1).

Table 1. Blocking views.

| view                   | how it matches                                                                                                                                                                           | what it catches                                                                                 |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Token view</b>      | IDF-weighted token overlap on name + address, searched both ways (each S1's top 40 records, each record's top 8 S1). A pair is kept if it ranks in the S1's top 15 or the record's top 4 | most copies. A (house number, street word) key that skips street types handles French addresses |
| <b>Names-only view</b> | character 4-grams; single-token names of 8+ letters                                                                                                                                      | empty or short addresses, garbles, domains and handles                                          |
| <b>Repairs</b>         | domain names split into the country's words; OCR digits in names fixed; an Indic→Latin dictionary (693 entries) learned from training pairs                                              | web-style names, OCR errors, transliterated names                                               |

**Candidate pairs generated.** Retrieval yields **58.4M test pairs** (66.4M train). A stage-0/1 gradient-boosted scorer then keeps the pairs with **p1 ≥ 0.02**, plus each record's **two highest-scoring S1**, and acronym joins are added. That leaves **6,410,308 candidates, or 3.70 per test S1** (`candidate_pairs.tsv`).

**How you ensured true matches were not lost.** We **measured blocking recall on the local holdout** rather than assuming it. **99.1%** of true pairs are candidates; the 16.5k misses (of about 1.9M) are mostly empty-address copies whose name changed. Tighter candidate cuts lost holdout F0.5 (top-1 per record  $-0.000106$ ;  $p1 \geq 0.05$   $-0.000041$ ), so we kept the more generous cut.

## 4. Matching Model

*Four XGBoost stages score each pair, cross-encoders read the uncertain ones, and a per-S1 decision picks the final set.*

**Features used.** All statistics are fitted per country; there is no country feature. Table 2 groups the features by the evidence they measure.

Table 2. Feature groups.

| group                             | main features                                                                                                                                                                                     | why they matter                                                        |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Name</b>                       | fuzzy ratios (token-sort, token-set, partial, Jaro-Winkler); token Jaccard and containment; IDF-weighted cosine; legal-form agreement; counts of unexplained tokens; learned look-alike word odds | copies keep the name's rare words, while decoys swap one business word |
| <b>Address &amp; house number</b> | house-number equality, nudges ( $\pm 1-10$ ), digit edits and suffixes; street overlap; city and state; whole-address similarity; empty and PO-box flags                                          | the address separates businesses that share a name                     |
| <b>Context</b>                    | rank in each blocking view; the gap to the S1's best candidate and to rival S1; name-collision counts; per-S1 aggregates; cluster support; the cross-encoder score                                | a pair is judged against its competitors                               |

**Model type.** Four **XGBoost** stages, each trained out-of-fold over three S1 groups so that later stages learn from honest scores. **Stage 0** prunes about 80% of pairs cheaply. **Stage 1** gives the probability  $p1$ , which sets the candidate cut and the uncertain band. **Stage 2** adds the cross-encoder, per-S1 and cluster features and the French pseudo-labels, and is isotonic-calibrated. **Stage 3** re-scores each pair in the context of its S1's other candidates.

**The cross-encoders** read only the **uncertain band**,  $0.02 \leq p1 \leq 0.99$  (1.49M test pairs), because that's where reading the text pays off. Each is trained as three out-of-fold models on `name` ; `address` of both records, up to 96 tokens. Their z-scored scores are **averaged into one stage-2 feature**: separate scores let stage 2 extrapolate where the models disagree, which happens four times as often in France. Table 3 lists the models.

Table 3. Cross-encoders (band AUC on labelled local-holdout pairs; stage-1  $p1$  alone scores 0.930).

| model                            | size        | licence    | training       | band AUC      |
|----------------------------------|-------------|------------|----------------|---------------|
| multilingual-e5-small / e5-large | 118M / 560M | MIT        | full fine-tune | 0.939 (large) |
| bge-reranker-v2-m3               | 568M        | Apache-2.0 | full fine-tune | 0.942         |

| model             | size | licence    | training                                      | band AUC     |
|-------------------|------|------------|-----------------------------------------------|--------------|
| Qwen2.5-1.5B      | 1.5B | Apache-2.0 | LoRA                                          | 0.938        |
| <b>Qwen2.5-7B</b> | 7.6B | Apache-2.0 | LoRA (r 16, $\alpha$ 32); 3 × H100, about 2 h | <b>0.944</b> |

French self-trained versions of e5-large and bge are also in the mix, and e5-base (278M, MIT) served only the first French teacher. Notably, the **7B is no more accurate** than a self-trained e5-large (both 0.944). Its value is **diversity**: it counts twice in the US/India mix, and it gives an **independent reading** of confident pairs in the re-check.

**Threshold selection method.** There is no single threshold. The calibrated stage-3 probabilities feed an **expected-F0.5 set selection**: a dynamic programme picks, for each S1, the prefix of its candidates with the highest expected F0.5, with settings tuned on the local holdout (a logit shift of +0.2, −0.3 for records contested by four or more S1, and 0.01 expected missed match). Each record goes to its highest-scoring S1. **Rules** then add acronym joins and per-source caps. For France they also apply an exact-address copy rule (99.99% precise on US/India), a cross-commune drop, a look-alike word-swap drop, and the same set selection on France's own probabilities. Finally, the **7B re-check** drops confident pairs with logit < −6. That cut-off gave the largest local-holdout gain (+0.000037), positive in both halves and in 99.7% of random subsets.

## 5. Results & Error Analysis

*US and India are close to solved. What remains is recall on empty-address copies, and precision on French decoys.*

**F\_0.5 Score (macro).** Table 4 gives the scores. Figure 2 shows how the leaderboard score was built up, one submission at a time.

Table 4. Macro F0.5.

| evaluation                                                           | macro F0.5                                                      |
|----------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>public leaderboard</b> (official; US + India + France)            | <b>0.990879</b>                                                 |
| local validation: our labelled holdout, US + India only (549,699 S1) | 0.9913 (US 0.9911, India 0.9916); precision 99.9%, recall 97.5% |

The local score is higher because France, which has no labels, cannot be part of it; most of the gap is France. The private leaderboard, which decides the final ranking, had not been published when we wrote this.

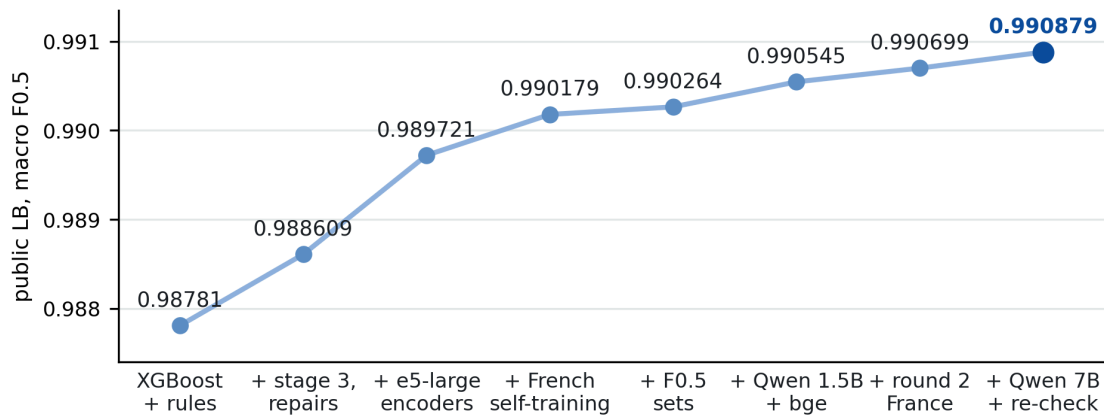


Figure 2. Public leaderboard score after each submission step. The last point is the submission in this package.

The largest steps were the **e5-large cross-encoders** (+0.0011) and the **blocking repairs with stage 3** (+0.0008). **French self-training** added +0.00046. The last step, the **Qwen2.5-7B mix and re-check**, added +0.00018.

**Common false positives (wrong merges).** The costliest were **generic-name decoys in France**: the same generic name and house number on a different street (`lille ecole sarl | 42 rue gutenber` vs ... | `42 q. du wault`). On labelled data this pattern is 0.5% true where our model rejects it and 99.7% true where it accepts it, so string features can't separate the cases; the 7B can. Removing its 840 French rejects gave an estimated **+0.00011** of our last +0.00018. Smaller groups are **business-word swaps** at the same address ("antenne danse" vs "antenne gaz"), **near-identical short names** ("osd" vs "otd"), and **chain branches** that share one name.

**Common false negatives (missed matches).** With 99.9% precision, the loss is mostly **recall**: 69% comes from partially missed S1 and 20% from S1 missed entirely. The missed copies are mostly **empty-address records with changed names**, which are either never retrieved (16.5k) or scored near zero (15.2k), plus **heavy garbles** such as the French "EHPAD" typed as "ehpvd". Every recall rule we tried was only 17–71% precise, below the ~75% an addition needs to raise F0.5, so **we added none**.

## 6. Conclusion

A **precision-first pipeline** with **calibrated, set-level decisions** brought US and India to 0.9913 on our local validation, and **guarded self-training** carried France most of the way without a single label, for **0.990879** on the public leaderboard. An **independent 7B re-check** of the predictions that looked certain found decoys that feature models accept. We learned two things: self-training paid for **two rounds**, and without labels any probability-based estimate needs a **second, independent judge**, because it inherits the model's blind spots.

## Appendix

### A. Code Artefacts

`code/business_entity_resolution/` regenerates both output files from the raw train/test TSVs. Its `README.md` has the exact commands, run times and checks.

| path          | contents                                                                                                                   |
|---------------|----------------------------------------------------------------------------------------------------------------------------|
| reproduce.sh  | the entry point. <code>VARIANT=compositeB</code> (the default) runs <code>src/box/compositeB.sh</code>                     |
| src/ber/      | the team package: I/O, normalisation, blocking, context features, evaluation (holdout split, macro F0.5, paired bootstrap) |
| src/model_v1/ | the XGBoost stages, cross-encoder training, self-training labels, the decision and the rules, the France block             |
| src/box/      | Qwen2.5-7B training, the 7B re-check, per-country composition                                                              |

**Run:** `pip install -r requirements.txt`, `pip install -e .`, then `VARIANT=compositeB bash reproduce.sh`. **Hardware:** 24+ CPU cores and 32 GB RAM for the XGBoost stages, one H100 80 GB for the cross-encoders, and three 80 GB GPUs for Qwen2.5-7B (about 2 hours). **Software:** Python 3.12–3.13, torch 2.11, transformers 5.17, peft 0.21, xgboost 3.2.0; Qwen2.5-7B revision d149729... **Reproducibility:** seeds are fixed, and every artifact records its command and git commit. GPU training is not bit-identical across machines (rebuilding an earlier model on other hardware gave 0.991261, against 0.991246), so a rerun lands within about  $\pm 0.0001$ ; `output/` holds the exact submitted bytes.

## B. Additional Results

Table B1. Ablations (local validation: labelled US/India holdout, paired bootstrap).

| change                                            | effect on macro F0.5                    |
|---------------------------------------------------|-----------------------------------------|
| expected-F0.5 set selection vs a tuned threshold  | +0.000048 [+0.000007, +0.000091]        |
| Qwen2.5-7B counted twice in the cross-encoder mix | India +0.000066 (P 0.998), US +0.000026 |
| 7B re-check (drop logit < -6)                     | +0.000037, both halves positive         |
| tighter candidate cut (top-1 / $p_1 \geq 0.05$ )  | -0.000106 / -0.000041                   |

Table B2. The 7B re-check on predictions with  $p_1 > 0.99$  (a labelled 34% sample of the local holdout, and the French test predictions).

| 7B logit     | holdout predictions | truly a match | French predictions |
|--------------|---------------------|---------------|--------------------|
| below -6     | 24                  | 8.3%          | 859                |
| -6 to -2     | 189                 | 92.6%         | 530                |
| -2 and above | 602,949             | > 99.9%       | 785,987            |

The 7B rejects **0.10% of French** predictions, against 0.004–0.007% of US/India ones, and it does so at the same rate in every third of French S1, including the third it never saw labels for. So **the re-check is not leaking its training labels**.

**Leaderboard controls.** Composite B with the round-3 French model but without the French decision layers scored 0.990833 (−0.000046). Extending the French 7B drops below −6 scored 0.990875, a tie, so the −6 cut-off was already right.

**What did not work** (measured, then dropped):

- **recall rules**, 17–71% precise;
- **a learned blend** of all six cross-encoder scores as the decision score: band AUC 0.9575, but –0.001 F0.5;
- **mining extra drops** with a decision tree: +0.000033 on one half, –0.000015 on the other;
- **copy-count tie-breaking**: –0.00057;
- **renormalising French probabilities** per record: –0.000007 to –0.000137;
- **other encoders**: Qwen3-4B scored below the 7B, and mDeBERTa-v3 and gte-multilingual failed to train or run.